

C Thomas, PhD

(they/them)

[Website](#) [LinkedIn](#) [GitHub](#) [Scholar](#) [Email](#) [SF Bay Area](#)

Professional Summary

- **AI R&D Leader bridging frontier science and real-world constraints.** Open to Head/Director of Applied AI Research, or high-impact research leadership focused on frontier model training, evaluation, and interpretability.
- **Expert in bridging research and engineering cultures,** aligning technical and executive teams to achieve high-stakes milestones across academia, startups, and enterprise. Consistently identify gaps and build new teams to solve them.
- **Built and lead an applied AI R&D function** of 25+ contributors across 6 research threads, translating research into standard ML training frameworks used in hundreds of critical global deployments.
- **Cut operational costs by 60% and accelerated model deployment from weeks to hours** by bringing engineering and science into one team, grounding work in factory realities, and driving end-user adoption.
- **Bring experience navigating startup growth and scientific research,** I lead with curiosity and direct communication, and push for an active-discussion environment with strong opinions, loosely held.

Skills & Tools

R&D Leadership:	Research-to-Production Translation, Cross-Functional Strategy, Team Vision & Engagement, Academic Partnerships, Mentorship
AI/ML Methods:	LLM Pretraining, Computer Vision & VLMs, Domain Adaptation, LoRA Fine-Tuning, Synthetic Data Augmentation, Model Evaluation & Benchmarking
Engineering & Scale:	Systems Engineering, MLOps Infrastructure
Tools:	Python, PyTorch, Hugging Face, LLM Foundry, Detectron2, OpenMMLab

Experience

Apple Cupertino, CA

Applied Research Manager, Machine Learning

2022 – current

- **Built and lead the applied ML research function,** standardizing core training libraries that enable 20+ MLEs and 5+ cross-functional teams to leverage research breakthroughs across hundreds of global deployments.
- **Direct a seven-figure portfolio of internal and external PIs across 6 parallel research threads,** responsible for compute allocation, collaboration budget, and technical gate management for an ecosystem of 25+ contributors.
- **Fostered career growth over 5+ years of ML leadership,** leading teams of researchers and engineers, mentoring students and guiding fellow tech leads to clarify intended impact, success metrics, and early progress signals.
- **Demonstrated impact of closed-loop R&D strategy** including reduced manufacturing-specific training data requirement for synthetic defect generation from 10,000 to just 5 images and cut overall annotation costs by 60%.
- **Engineered the evaluation pipeline for a novel memory-augmented LLM (ICLR 2026);** authored a patch to LLM Foundry, built the memory-retrieval routing logic, and executed end-to-end evaluations comparing baseline and augmented architectures.

Machine Learning Engineer

2020 – 2022

- **Reduced urgent model deployment time by 80%** by implementing MLOps infrastructure; bridged historical divides between ML, Data Engineering, and Software teams to build a unified data management platform.
- **Trained production computer vision models** for deployment on factory lines, building monitoring systems to detect model drift and core vulnerabilities prior to customer impact.
- **Established engineering standards** and mentorship to junior ICs, interns and student collaborators in collaborative software development and rigorous experimental data interpretation.

Tua Fertility Oakland, CA

Founder & CEO

2019 - 2020

- **Designed, patented and prototyped a medical device,** executing end-to-end mechanical design (CAD/SolidWorks) and rapid iterations to proof of concept as a resident at the Autodesk Technology Center
- **Drove technical vision and business strategy,** pitching to investors and translating complex engineering roadmaps into compelling narratives.

Rigetti Computing Berkeley, CA

Senior Research Engineer & Tech Lead

2017 - 2019

- **Navigated early-stage startup ambiguity to drive critical technical breakthroughs** by aligning teams and sub-systems to stabilize and optimize complex, non-linear computing architectures.
- **Built robust R&D software** to calibrate and run quantum gate operations, analyzing large datasets and distilling results into shared tooling for research teams.
- **Influenced executive technical roadmaps** by ruthlessly prioritizing actionable R&D threads, translating complex engineering progress into actionable milestones that shifted company timelines and targets.

University of California, Berkeley Berkeley, CA

Doctoral Research Fellow

2010 - 2017

- **Awarded the highly competitive DOE SCGF (accepted) and NSF (declined) Fellowships.**
- **Designed and built high-precision optics, hardware and electronic systems** to trap individual atoms in lattices of laser light. Drove quantum phase transition and explained dynamics with first-principles mean-field model ([Phys. Rev. Lett., Editor's Suggestion](#)).
- **Resolved a contested phenomenon in quantum literature** by developing a predictive white-box model that provided a clear explanation for unexpected system behavior ([Phys. Rev. A](#)).

Selected Publications

Pretraining with hierarchical memories: separating long-tail and common knowledge.

H. Pouransari, D. Grangier, [C. Thomas](#), M. Kirchhof, O. Tuzel. *ICLR 2026*. [[arXiv](#)]

Rethinking Semi-Supervised Domain Adaptation for Semantic Segmentation in the Era of Foundation Models. J. Kurien*, B. Balaji*, K.H. Lai, P. G. Vela, [C. Thomas](#), A. Wong, S. Rambhatla. *CVPR 2025 (FGVC Workshop)*.

LangDA: Language-guided Domain Adaptive Semantic Segmentation. C. Liu, S. Hossain, [C. Thomas](#), K.H. Lai, R. Vemulapalli, S. Rambhatla, A. Wong. *NeurIPS 2024 (AFM Workshop)*. [[NeurIPS](#)] [[arxiv](#)]

Synth4Seg: Learning Defect Data Synthesis for Defect Segmentation using Bi-level Optimization. S. Mou, R. Vemulapalli, S. Li, Y. Liu, [C. Thomas](#), M. Cao, H. Bai, O. Tuzel, P. Huang, J. Shan, J. Shi. *IEEE T-ASE 2024*. [[arXiv](#)]

Service & Leadership

Academic Conference Leadership

Co-organizer, [4th Workshop on Vision-based Industrial Inspection](#), CVPR 2026

Invited Speaker, "Applied Research in Industrial AI", [3rd Workshop](#), ICCV 2025

Co-organizer & Panelist, Data-efficient defect segmentation, [2nd Workshop](#), ECCV 2024

Career Mentorship

Committee Member, Employee Career Development & Rotations, Apple Operations 2023 – Current

Career Mentor, 1:1 guidance for recent graduates entering the workforce 2023 – Current

Community

Impact Bay Area is a nonprofit in the SF Bay Area that trains women and girls to set and hold boundaries with verbal and physical self-defense skills.

Board Secretary 2023 – 2025

Assistant Instructor 2017 – 2025

Education

PhD **University of California at Berkeley**, Atomic and Optical Physics 2017

[Quantum simulation of triangular, honeycomb and kagome crystal structures using ultracold atoms in lattices of laser light](#)

MA **University of California at Berkeley**, Physics 2013

BA **Boston University**, Physics and Math 2010